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Data Augmentation for Object Detection

3 min
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Data Augmentation for Object Detection

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Example of Augmentation Improving Performance

30 min
- Quiz

Graded Quiz: Data Augmentation

Submitted

Graded Quiz: Data Augmentation

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Due Mar 10, 11:59 PM IST

Receive grade

To Pass 85% or higher

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In this week's lesson, you saw that the fastener detection model's mAP improved when applying data augmentation to the training data. To further explore the improvement:

- Create a new live script to evaluate both fastener detection models (the model trained without data augmentation and with data augmentation) using an IoU of 0.5 and a range of detection thresholds, including 0.05 and 0.1. Use `EvaluatingAugFastenerDetectorResults.mlx` as a reference, but **do not overwrite that file**.
- Compare the results to what you found earlier (using an IoU of 0.25).
- Answer the quiz questions below.

Try again

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1. Identify the mAP at a detection confidence threshold of 0.1 and IoU threshold of 0.5 for the original and augmented models. Compare these results to the earlier ones where you had an IoU value of 0.25. Which of the statements below is true?

1 / 1 point

☐

Both models show a less than 0.08 decrease in mAP when increasing the IoU threshold.

☐

Both models show a greater than 0.08 decrease in mAP when increasing the IoU threshold.☒☐

☒

Correct

The model trained with augmented data has a high mAP at both IoU thresholds.

2. Look at the AP of the Nail and Screw classes using an IoU threshold of 0.5 and **no** data augmentation. Which statement below is true?

1 / 1 point

☒

The AP of the Nail class is much higher than the Screw class.

☐

The AP of the Screw class is much higher than the Nail class.☐

☒

Correct

Regardless of the detection threshold, the Nail class outperforms the Screw class.

3. Look at the AP of the Nail and Screw classes using an IoU threshold of 0.5 **with** data augmentation. Which statement below is true?

1 / 1 point

☐

The AP of the Nail class is much higher than the Screw class.

☐

The AP of the Screw class is much higher than the Nail class.☒

☒

Correct

The model trained with augmentation has similar performance for both these classes.

4. Look at the Nail and Screw classes for the model with **no** data augmentation. Compare the APs at a detection threshold of 0.05 to the ones you calculated earlier (with an IoU of 0.25). What statement below is the most likely reason for the changes in AP?

1 / 1 point

☐

The AP is similar for both IoU thresholds, indicating the location prediction is not an issue.

☒

The AP decreases because the model is struggling to predict accurate bounding boxes.☐

☒

Correct

Yes - these classes show a big decrease in AP when the IoU increases, meaning location accuracy is an issue.

5. In what ways can data augmentation improve model performance for object detection models?

1 / 1 point

☐

Reducing the number of false positives.

☐

Reducing the confusion between classes, like detecting a screw as a nail.☐☐☒

☒

Correct

Depending on the application, some options may be more important than others. For the fasteners, the detection accuracy was very important.

The following questions are general and not specific to the fastener models.

6. What is the main difference when training classification and object detection models with data augmentation?

1 / 1 point

☐

Object detection models have fewer transformation options to ensure bounding boxes remain around the objects.

☐

To use augmentation for object detection, you must save and label the augmented images.☒

☒

Correct

Yes, the bounding boxes must be adjusted for object detection

7. Which scenario below is a good fit for data augmentation?

1 / 1 point

☐

You have images acquired during daylight hours, but you also need images taken at night, so you use augmentation.

☐

You have images of two-door cars, but you also need images of vans and commercial trucks, so you use augmentation.☒

☒

Correct

Data augmentation is good option to add greater variety to your data.

